

**“Quantitative Estimation and Comparative Study of Caffeine
Content in Selected Beverages Using UV Spectrophotometry”**

PROJECT WORK REPORT SUBMITTED IN THE PARTIAL

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RESEARCH GUIDE

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CERTIFICATE

This is to certify that the work presented in the practice school report entitled Practice School of Quality Assurance submitted in partial fulfilment of the requirements for the degree of **Bachelor of Pharmacy**, under the Faculty of **Science and Technology**, **Savitribai Phule Pune University** has been carried out by Mr Sumit Kamble, Mr Yuvraj Jagtap, Mr Sushrut Randive, Ms Sayali Renge, Mr Samarth Sangale under my supervision. All the materials and services from other resources have been duly acknowledge in the project.

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ABBREVIATION

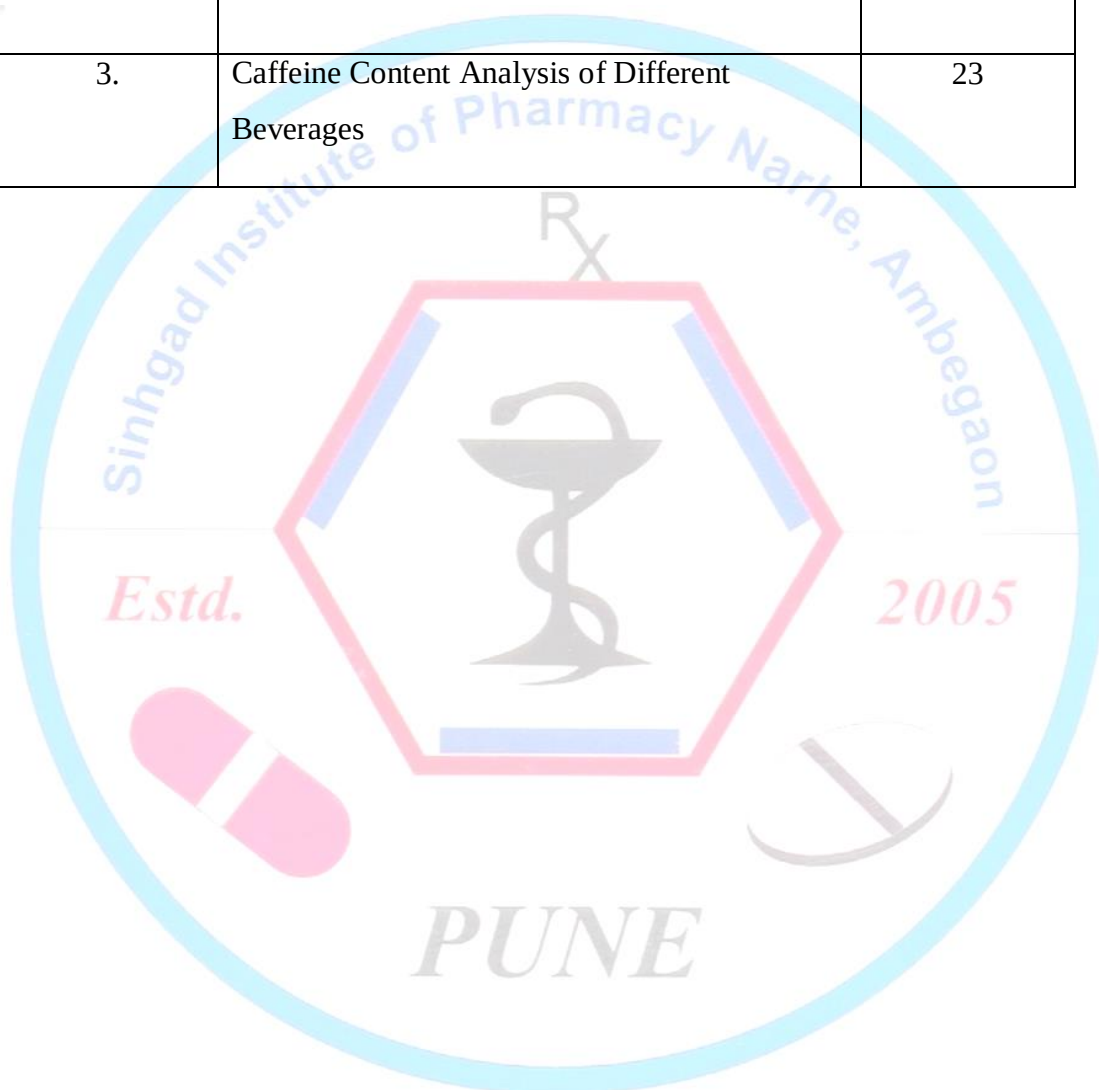
$^{\circ}\text{C}$	Degree Celsius
%	Percentage
ml	Mili litre
gm	Gram
Min	Minute
HCL	Hydrochloric Acid
TLC	Thin Layer Chromatography
RF	Retention Factor
AUC	Area Under Curve

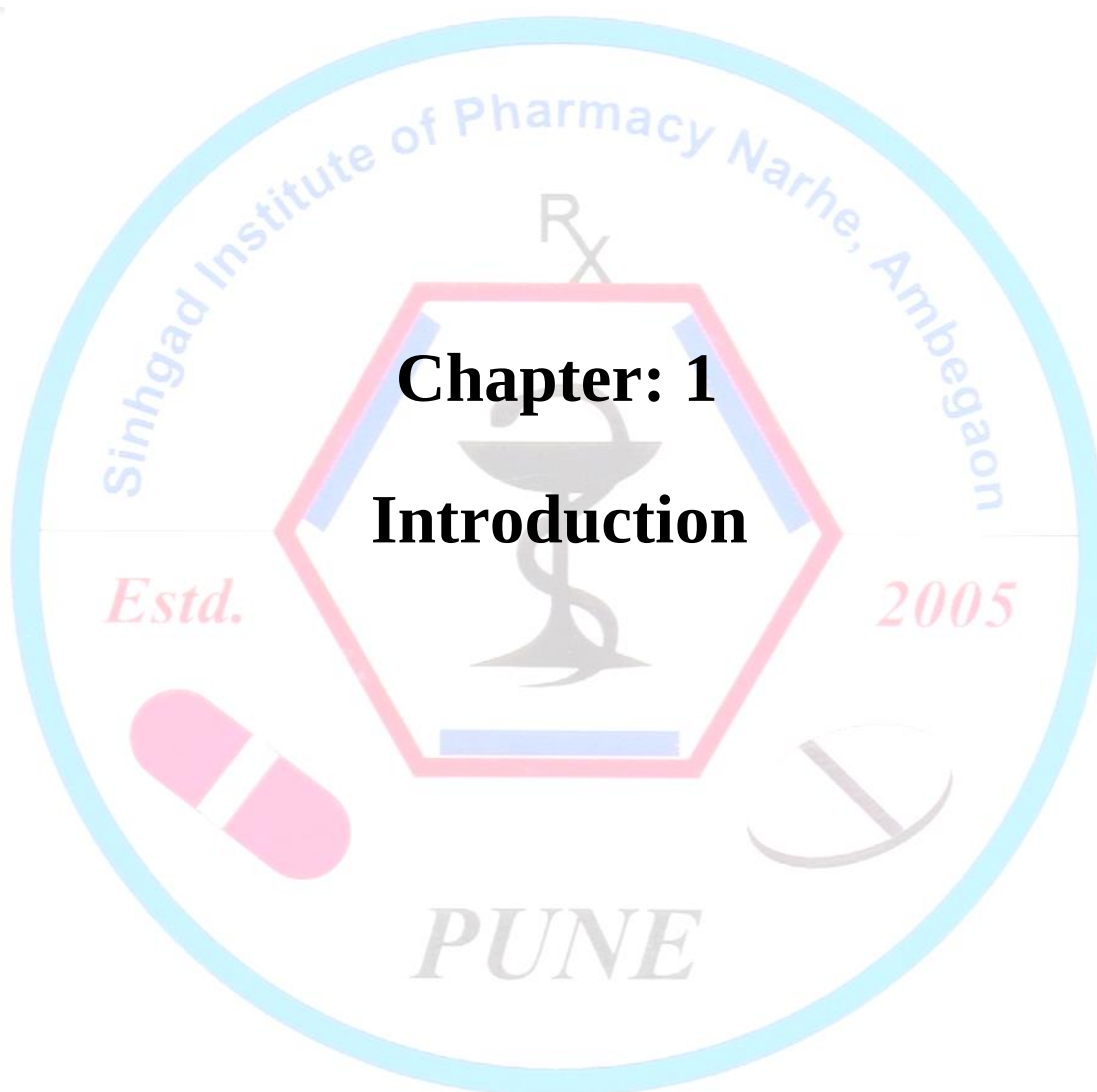
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Chapter: 1

Introduction

UV SPECTROMETRY

Analytical chemistry uses light absorption to characterize and quantify compounds. UV/Vis spectroscopy is a method based on how light is absorbed by an unknown material or sample. Here, electromagnetic waves of different visible wavelengths are used to illuminate the sample. [1]

The greater the number of molecules capable of absorbing light of a given wavelength, the greater the extent of light absorption. Furthermore, the more effectively a molecule absorbs light of a given wavelength, the greater the extent of light absorption.[7]

In the visible and adjacent (near-, mid-UV, and near-infrared) ranges, specifically 190-2500 nm (0.5-6.5 eV), this technique measures optical transmittance/reflectance, or the intensity ratios of the transmitted/reflected light from a material to the incident monochromatic light. The spectra, which provide details about the electronic excitation states in solids and molecules, are often found by scanning the monochromator.[2]

Beers Lambert Law:

$$A = \epsilon bc$$

Where:

A = Absorbance

ϵ = Molar absorptivity

b = Path length of the cuvette

c = Concentration of the solution.[3]

Parts of UV Spectrophotometer:

Light

There are two types of devices that provide incident light that meets the requirements: thermal radiation light sources and gas discharge light sources. [4]

Monochromator

Radiation is dispersed based on wavelength using a monochromator. An entrance slit, a dispersing element, and an exit slit are the fundamental components of a monochromator. The incoming beam of heterochromatic radiation is clearly defined by the entrance slit. While the exit slit permits the nominal wavelength along with a band of wavelengths on each side of it, the dispersing element breaks up the heterochromatic radiation into its component wavelengths. The nominal wavelength that passes through the exit slit is always changed by rotating the dispersing element.[6] A device that converts optical signals into electrical signals. When measuring

Detector

Absorbance, it does not directly measure the intensity of light passing through the absorption cell, but converts the intensity of light into an electrical signal for testing. This type of photoelectric conversion device is called a detector. Monitor The device that amplifies the signal output by the detector and displays it.[4]

Measurement of Absorption Intensity

It may be noted that the intensity of absorption is directly proportional to the transition probability. An allowed transition will have E_r value greater than 1000 while those having low transition probability will have its value less than 1000.[8]

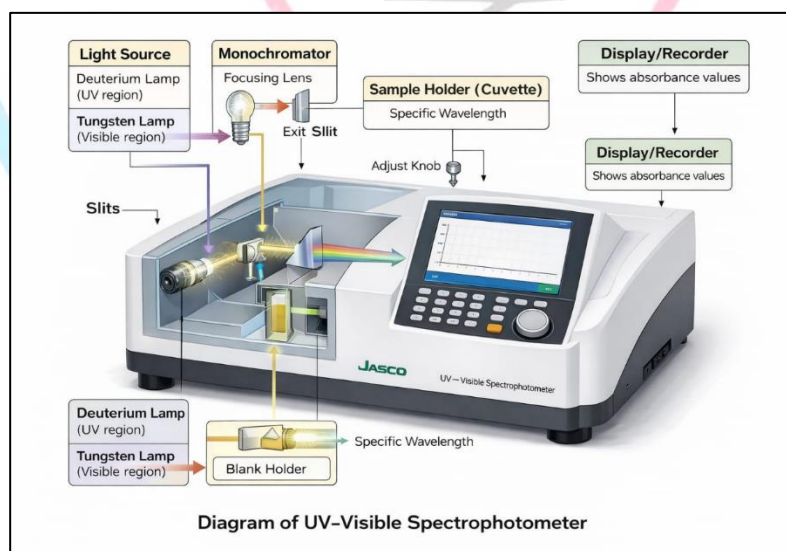


Fig.no: 1 – UV Visible Spectroscopy

Applications of UV Spectrophotometry

UV-Visible spectrophotometry is one of the most widely used analytical techniques in pharmaceutical, chemical, and biological sciences.

1. Analyzing pharmaceuticals:

One use for UV spectroscopy in drug development is the screening and characterisation of possible medicinal molecules [5]

2. Drug Identification

Every medication has an own UV absorption spectrum. It is feasible to ascertain whether the drug's structure matches the standard by comparing the UV absorption curves of the sample and the standard.[4]

3. Biochemical and Biomedical Research:

UV spectrophotometers are used to study biomolecules such as proteins and nucleic acids. It aids in the identification, measurement, and monitoring of these molecules' structural changes.[3]

4. Beverage analysis:

This technique is used to identify specific compounds present in drinks by analyzing bacterial cultures. For example, they can be easily recognized in wine for quality control purposes using UV absorption by matching what are known as the peak absorption wavelengths [5]

Steps involved in UV Spectrophotometry

1. Generation of UV light

A UV light source (usually deuterium lamp) produces radiation in the UV region (200–400 nm).

2. Selection of wavelength

The monochromator selects a specific wavelength of UV light required for analysis.

3. Passing through sample

The selected UV light passes through the sample solution placed in a quartz cuvette.

4. Absorption of light

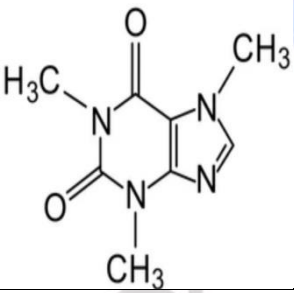
The analyte (e.g., caffeine) absorbs UV light at a specific wavelength (λ_{max}), causing electronic transitions.

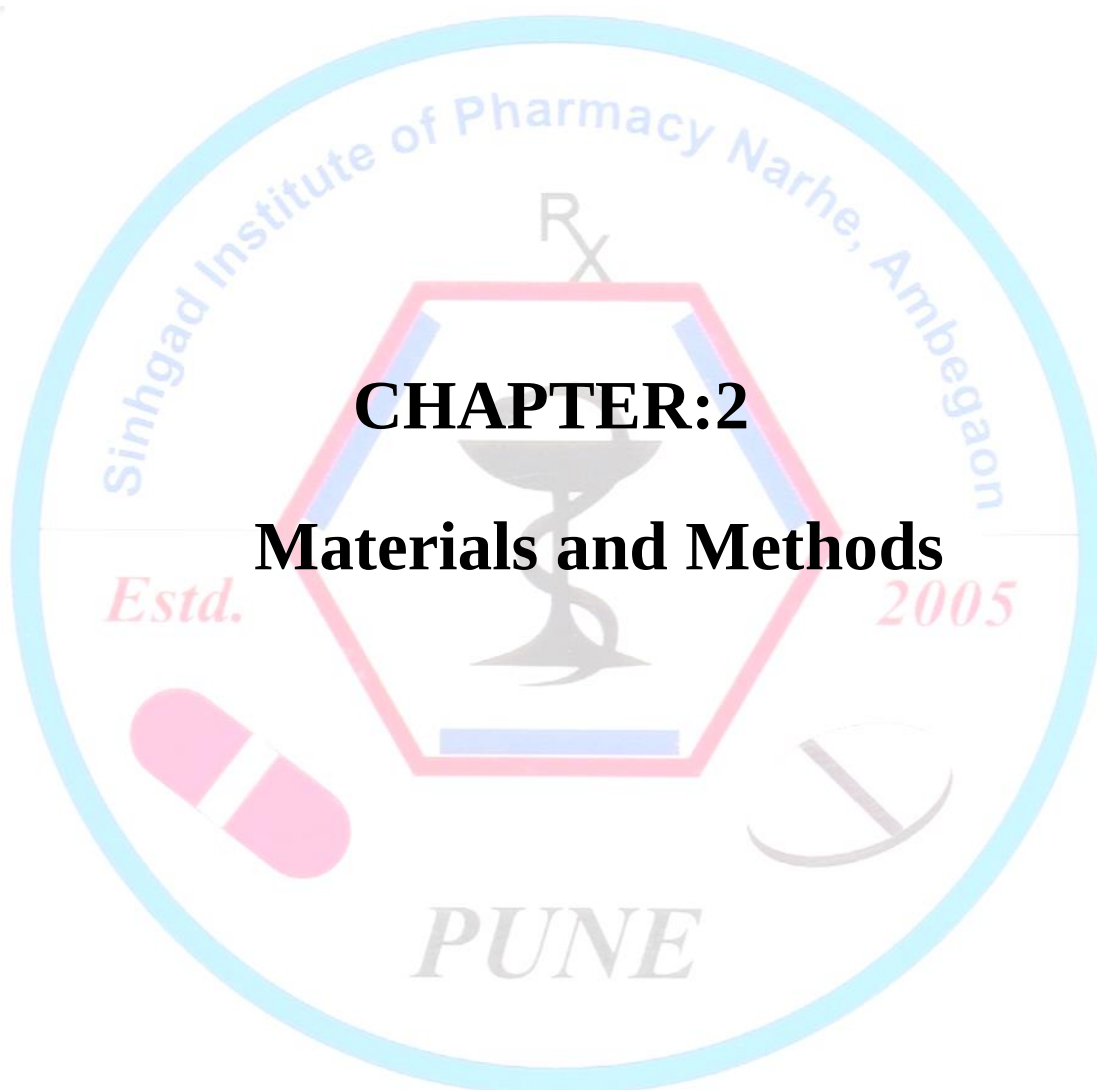
5. Measurement of absorbance

The instrument calculates absorbance using Beer-Lambert law.

Drug Profile:

Table 1 – Drug Profile of Caffeine

Structure	
Mol. Formula	$\text{C}_8\text{H}_{10}\text{N}_4\text{O}_2$
Mol. Weight	194.19g/mol
Melting Point	235–238 °C
Solubility	Sparingly soluble in water; freely soluble in boiling water; freely soluble in chloroform; slightly soluble in ethanol.
Category	CNS Stimulant, Analeptic
Drug class	Methylxanthine
Description	Caffeine is a naturally occurring alkaloid used to improve mental alertness and as an adjuvant in the treatment of migraines. It occurs as a white, odourless, bitter-tasting crystalline powder
Storage	Store in a well-closed container at controlled room temperature 20°C to 25°C (68°F to 77°F).



2.1 Chemicals and Reagents

All chemicals and reagents used in the present study were of analytical grade to ensure accuracy and reliability of results.

Caffeine standard (pure drug): Used as reference standard for calibration curve preparation.

Distilled water : Used as solvent for preparation of standard and sample solutions.

Commercial energy drinks: Various marketed beverages (e.g. Monster, Sting, etc.) were selected for caffeine estimation.

All solutions were freshly prepared to avoid degradation and ensure precision.

2.2 Instruments and Apparatus

The following instruments and apparatus were used during the experiment:

UV-Visible Spectrophotometer: Used to measure absorbance of solutions at selected wavelength (λ_{max}).

Quartz cuvettes (1 cm path length): Used for sample analysis as they allow UV light transmission.

Analytical balance: For accurate weighing of caffeine standard.

Volumetric flasks (10 ml, 50 ml, 100 ml): For preparation of standard and sample solutions.

Pipettes and micropipettes: For accurate measurement and dilution.

Beakers and measuring cylinders: For general laboratory use.

Sonicator (optional): To remove dissolved gases from samples.

All instruments were calibrated prior to use.

2.3 Description of Energy Drink Samples

Four different commercially available energy drinks were selected for analysis. These samples were procured from the local market.

Samples were selected based on popularity and availability.

Each sample was labelled with its declared caffeine content (mg per serving).

The samples were stored at room temperature and analyzed before the expiry date.

The purpose of selecting multiple samples was to compare the actual caffeine content with label claims and evaluate product quality.

2.4 Experimental Procedure

2.4.1 Determination of λ_{max}

A standard caffeine solution (10 $\mu\text{g/ml}$) was prepared.

The solution was scanned in the UV spectrophotometer over a wavelength range of 200–400 nm.

The wavelength at which maximum absorbance was observed was noted as λ_{max} .

The λ_{max} for caffeine was found to be approximately 272 nm.[13]

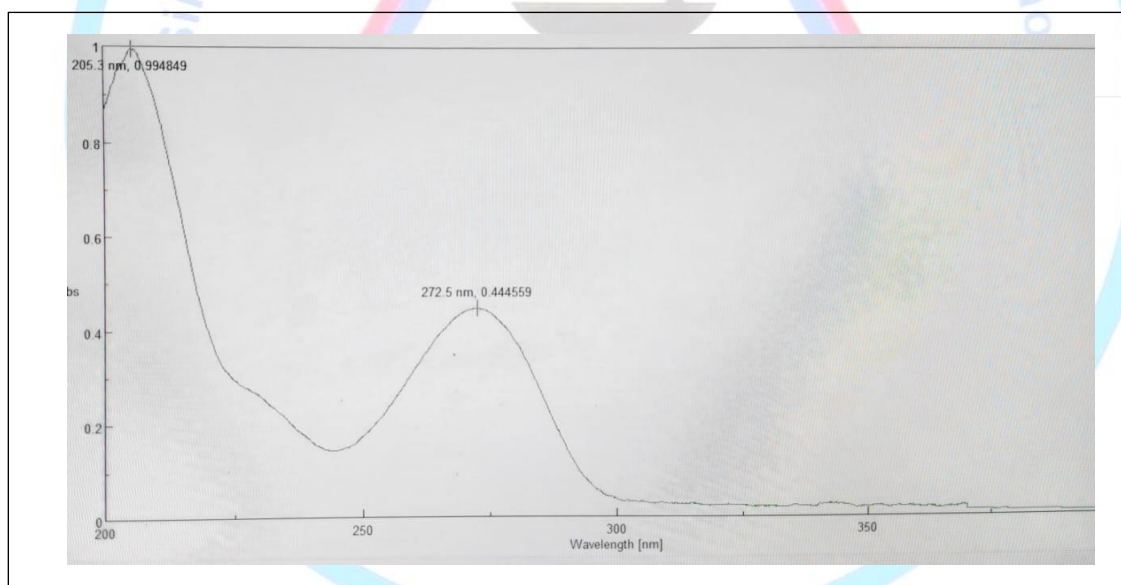


Fig.no. 2 Determination of λ_{max}

The maximum absorbance (λ_{max}) of caffeine was found at 272nm which was used for further analysis.

A = Absorbance

m = Slope = 0.0366 c = Intercept = 0.4756

2.4.2 Calibration Curve Preparation

Preparation of Standard Caffeine Solution

Accurately weighed 10 mg of caffeine was dissolved in 10 mL of distilled water to obtain a primary stock solution of concentration:

Concentration = 100 µg/ml

From this solution, **1 mL** was further diluted to **10 mL with distilled water** to prepare a secondary stock solution of concentration.

Concentration = 100 µg/ml

2.4.3 Preparation of Working Standard Solutions

From the secondary stock solution (100 µg/mL), aliquots of 1, 2, 3, 4, 5, and 6 mL were pipetted into separate 10 mL volumetric flasks and diluted up to the mark with distilled water to obtain working standard solutions of:

- 10 µg/mL
- 20 µg/mL
- 30 µg/mL
- 40 µg/mL
- 50 µg/mL
- 60 µg/mL

2.4.4 Absorbance of Standard Caffeine Solutions

The absorbance of the prepared standard solutions was measured at the determined λ_{max} using a UV spectrophotometer

Table 2 – Concentration and Absorbance

Concentration (µg/mL)	Absorbance
10	0.8191
20	1.1504
30	1.5771
40	2.0600
50	2.3928
60	2.5384

Calibration Curve

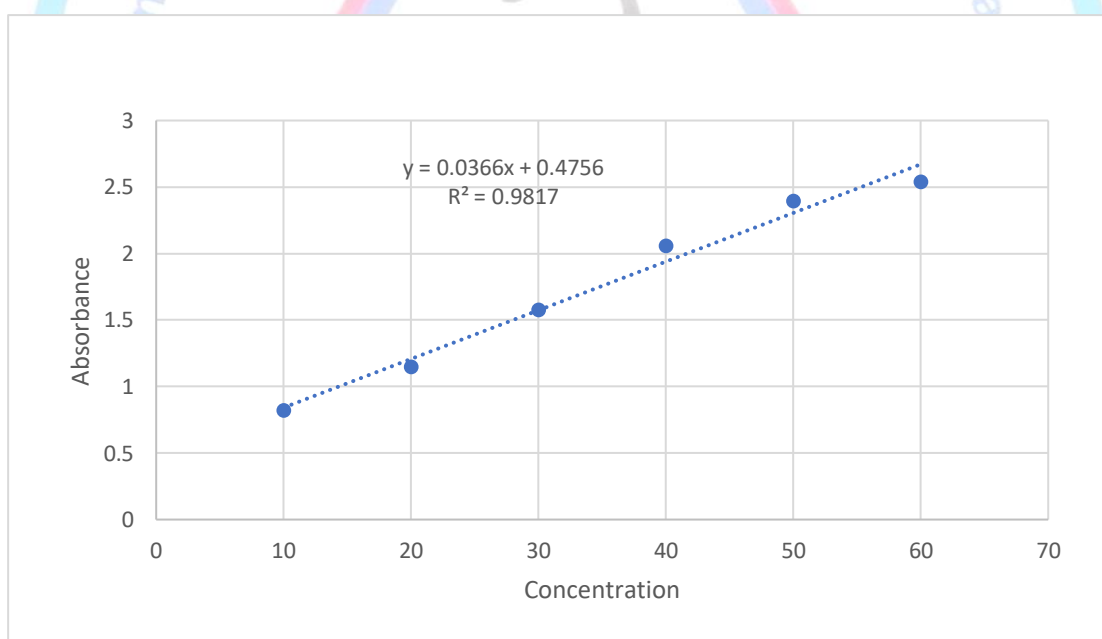


Fig.no. 3 – Calibration Curve of Caffeine

A calibration curve was plotted between concentration (µg/mL) on the X-axis and absorbance on the Y-axis. The calibration curve showed a linear relationship within the selected concentration range and followed **Beer-Lambert's law**. The regression equation obtained was:

$$Y = 0.0366 + 0.4756$$

Where:

- Absorbance
- x = Concentration ($\mu\text{g/mL}$)

The maximum absorbance (λ_{max}) of caffeine was found at 272nm which was used for further analysis

2.5 Absorbance of Samples

The concentration of caffeine was calculated using the regression equation obtained from the calibration curve

$$Y = mx + c$$

A = Absorbance

m = Slope = 0.0366

c = Intercept/Concentration = 0.4756

Rearranging: $X = (y - c) / m$

Confirmatory Test

Principle

A chromatographic method called thin-layer chromatography (TLC) is used to separate and identify chemicals based on how differently they migrate over a stationary phase while being influenced by a mobile phase.[12] [15]

The R_f (Retention factor) value of caffeine may be determined since it is a moderately polar molecule that travels along the stationary phase (silica gel) with the mobile phase (chloroform: methanol).[9]

Materials Used

- Stationary phase: **Silica gel TLC plate**
- Mobile phase: **Chloroform : Methanol (9:1)**
- Standard caffeine
- Beverage sample
- Capillary tubes
- Developing chamber UV chamber (254 nm) [14]

Procedure

1. A TLC plate coated with silica gel was taken and a baseline was drawn at about **1–2 cm from the bottom**.
2. The sample solution containing caffeine was applied as a small spot using a capillary tube.
3. The TLC plate was placed in a developing chamber containing the mobile phase (chloroform: methanol 9:1).
4. The solvent was allowed to travel up the plate.
5. The plate was removed when the solvent front reached an appropriate height and then dried.
6. The spot was visualized under **UV light (254 nm)**.
7. The distance travelled by the solute and solvent front was measured.¹¹

Observations

- Distance travelled by solute (caffeine) = **4 cm**
- Distance travelled by solvent front = **5.7 cm**



Fig.4 – TLC Plate

Calculation of Rf Value

Rf value is calculated using the formula:

$$R_f = \text{Distance Travelled by Solute} / \text{Distance Travelled by Solvent} [10]$$

$$= 4 / 5.7$$

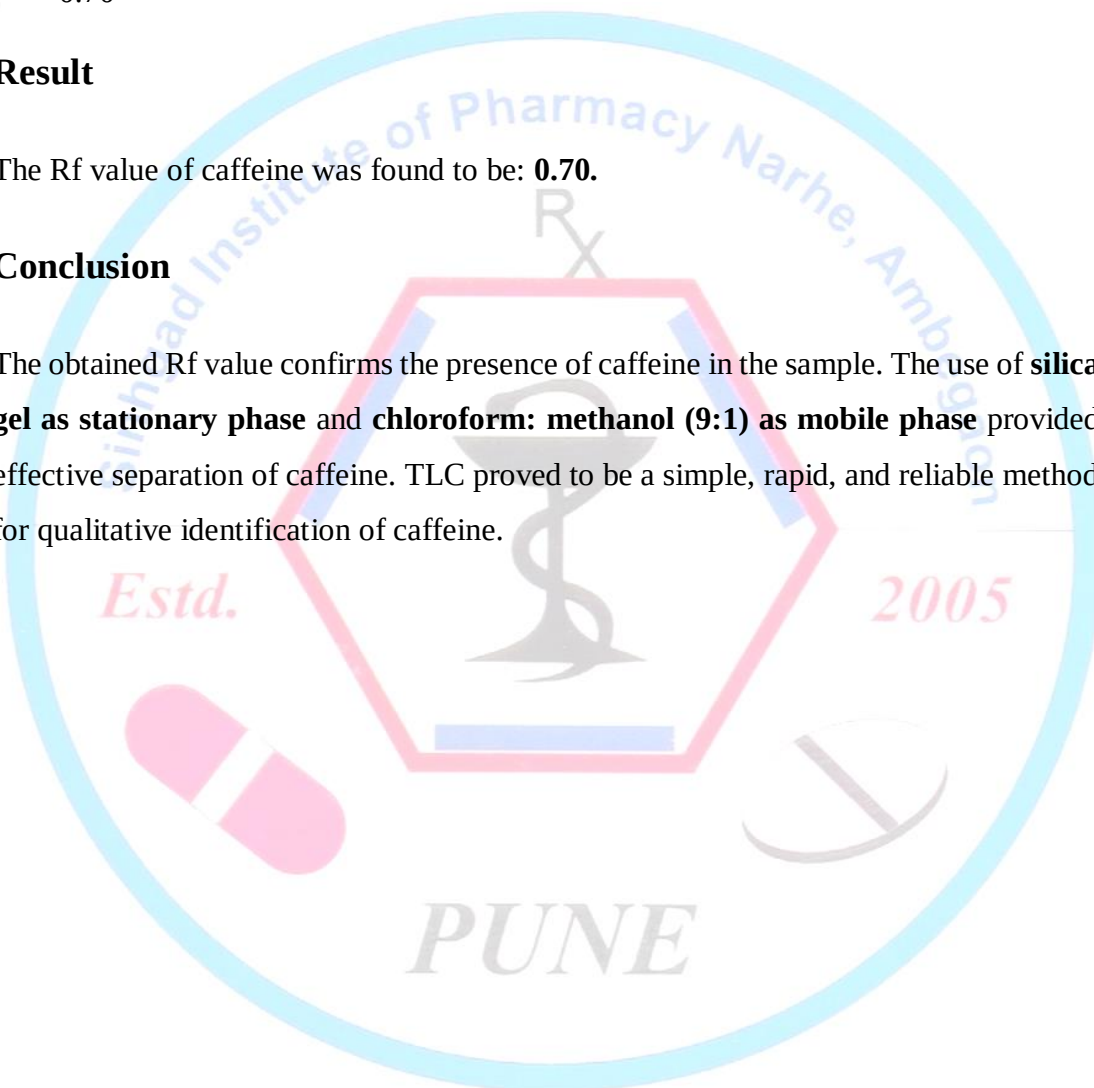
$$= 0.70$$

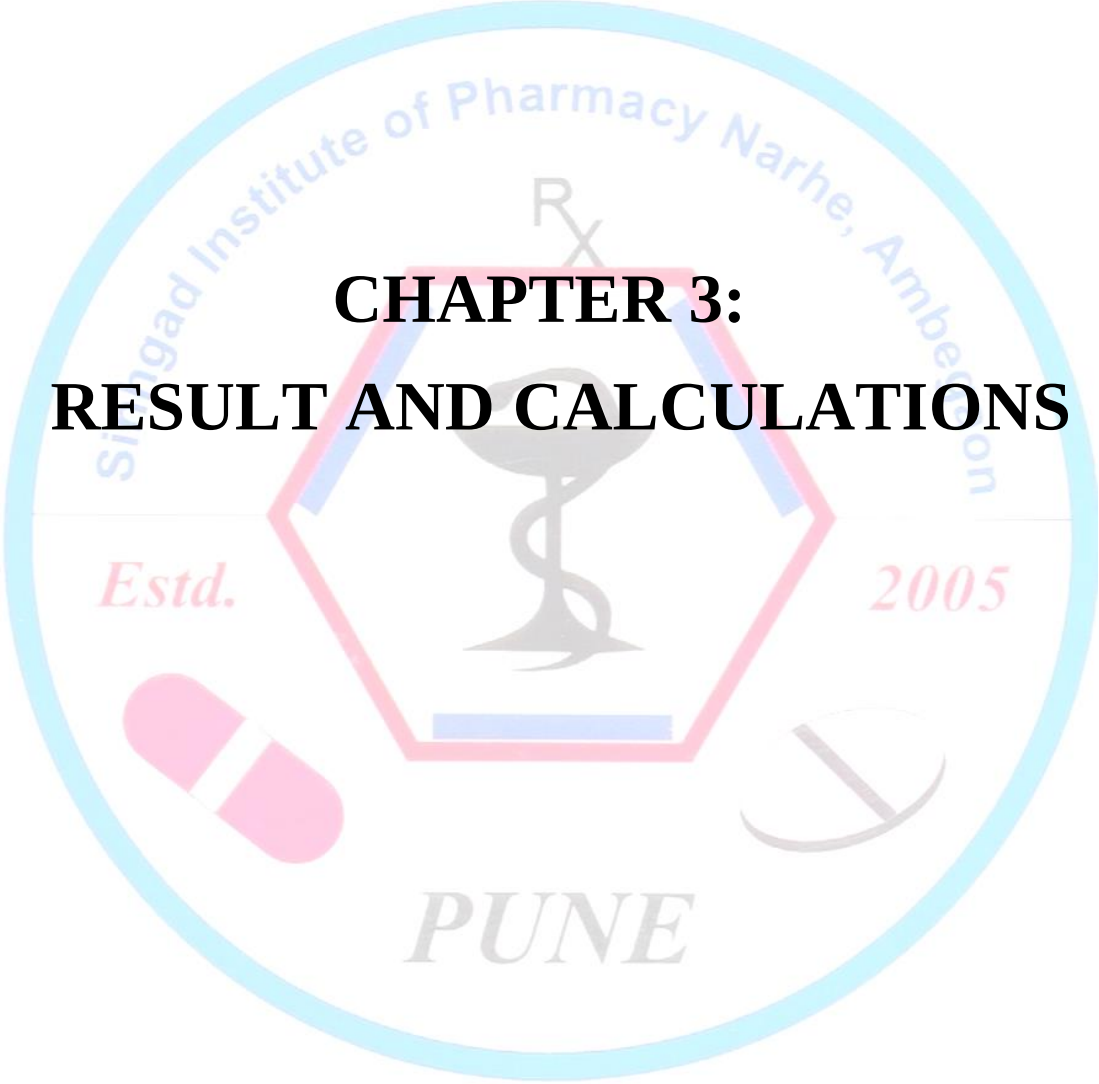
Result

The Rf value of caffeine was found to be: **0.70**.

Conclusion

The obtained Rf value confirms the presence of caffeine in the sample. The use of **silica gel as stationary phase** and **chloroform: methanol (9:1) as mobile phase** provided effective separation of caffeine. TLC proved to be a simple, rapid, and reliable method for qualitative identification of caffeine.



The logo of Singad Institute of Pharmacy Narhe, Ambernol is a circular emblem. It features a central caduceus (a staff with two snakes entwined and wings at the top) inside a hexagon. The hexagon has a pink border and a blue border. The text "Singad Institute of Pharmacy Narhe, Ambernol" is written in a light blue arc at the top. Below the hexagon, the word "PUNE" is written in a large, grey, serif font. To the left of the hexagon, the word "Estd." is written in a pink, italicized font. To the right, the year "2005" is written in a pink, italicized font. At the bottom left, there is a pink and white capsule, and at the bottom right, there is a white tablet with a diagonal line.

CHAPTER 3:

RESULT AND CALCULATIONS

Sample 1 - Sting

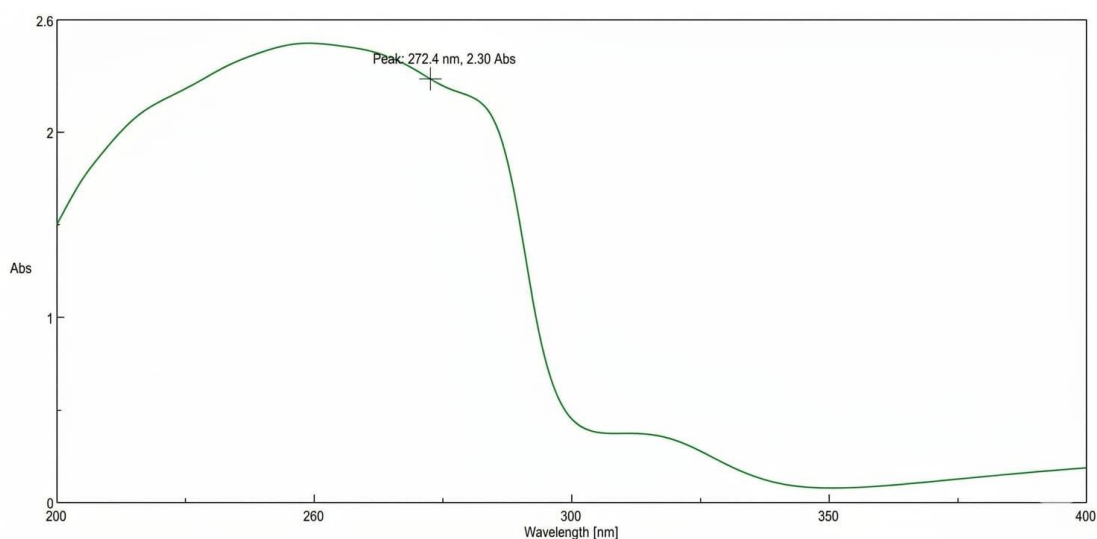


Fig.5 – Sting Spectra

Step 1: Substitution of Values

Absorbance of Sting drink = 2.30

$$X = 2.30 - 0.4756 / 0.0366$$

$$= 1.8244 / 0.0366$$

$$= 49.85 \text{ ug/ml}$$

Step 2: Applying Dilution Factor

Dilution factor = 5.75

Actual concentration = 49.85×5.75

$$= 286.64 \text{ ug/ml}$$

Step 3: Calculation for 250 mL Beverage

Total caffeine = 286.64×250

$$= 71660 \text{ ug}$$

$$= 71.66 \text{ mg}$$

The caffeine content in **250 mL of Sting drink** was found to be: **71.66 mg**



Fig.no.6-Sting

Sample 2 – Mountain Dew

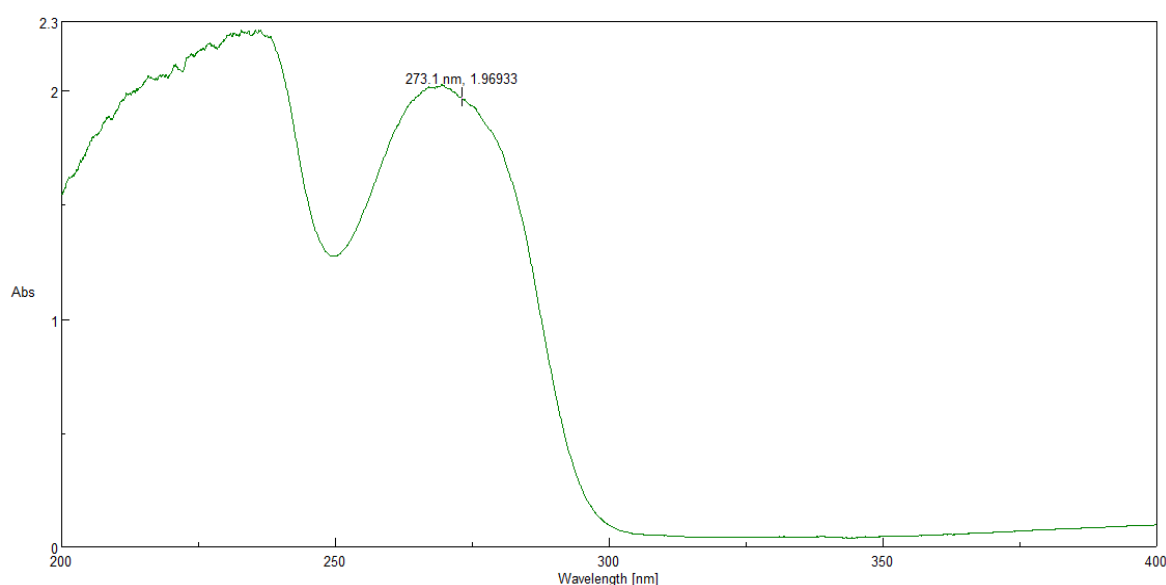


Fig.no. 7 – Mountain Dew Spectra

Step 1: Substitution of Values

Absorbance of Mountain Dew = 1.9

$$X = 1.9 - 0.4756 / 0.0366$$

$$= 1.4244 / 0.0366$$

$$= 38.93 \text{ ug / ml}$$

Step 2: Applying Dilution Factor

Dilution factor = 3.25

$$\text{Actual concentration} = 38.93 \times 3.25$$

$$= 126.52 \text{ ug/ml}$$

Step 3: Calculation for 400 mL Beverage

$$\text{Total caffeine} = 126.52 \times 400$$

$$= 50608 \text{ }\mu\text{g/ml}$$

$$= 50.61 \text{ ug/ml}$$

The caffeine content in **400 mL of Mountain Dew** was found to be: **50.61 mg**



Fig.no. 8-Mountain Dew

Sample 3 -Predator

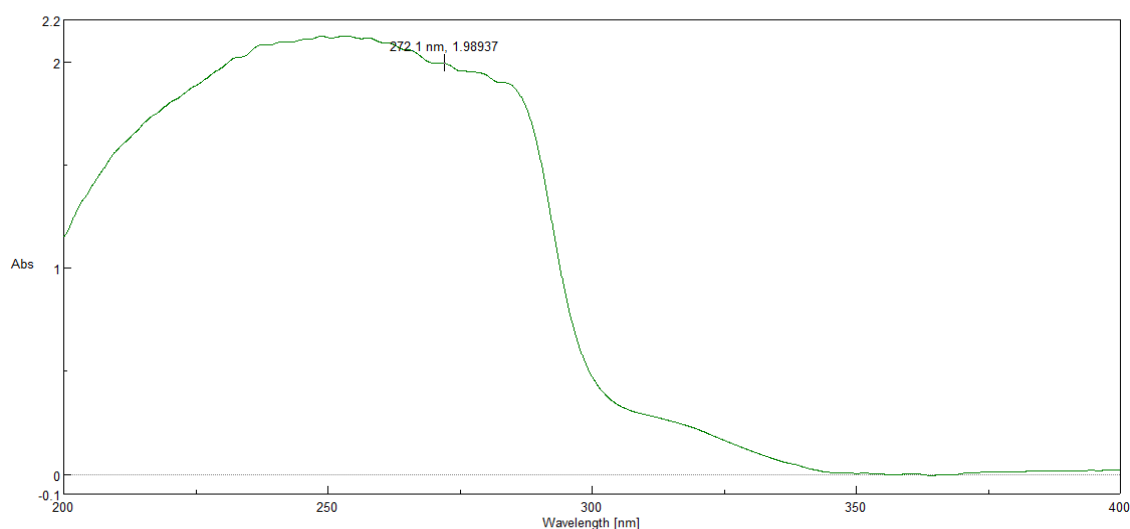


Fig.no. 9 – Predator Spectra

Step 1: Substitution of Values

Absorbance of Predator drink = 1.9

$$X = 1.9 - 0.4756/0.0366$$

$$X = 1.4244/0.0366$$

$$X = 38.93 \text{ ug/ml}$$

Step 2: Applying Dilution Factor

$$\text{Dilution factor} = 10/1.67 = 5.98 = 6$$

$$\text{Dilution factor} = 6$$

$$\begin{aligned} \text{Actual concentration} &= 38.93 \times 6 \\ &= 233.58 \text{ ug/ml} \end{aligned}$$

Step 3: Calculation for 300 mL Beverage

$$\text{Total caffeine} = 233.58 \times 300$$

$$= 70074 \text{ ug/ml}$$

$$= 70.07 \text{ mg}$$

The caffeine content in **300 mL of Predator drink** was found to be: **70.07 mg**



Fig.no.10 -Predator

Sample 4 – Monster

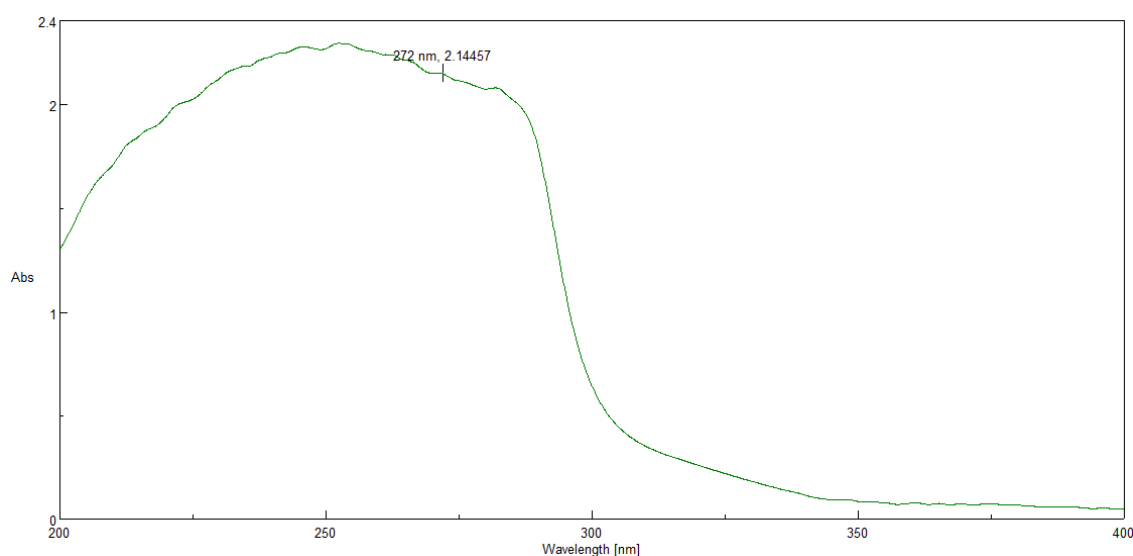


Fig. 11 - Monster Spectra

Step 1: Substitution of Values

Absorbance of Monster drink = 2.1

$$X = 2.1 - 0.4756/0.0366$$

$$= 1.6244 / 0.0366$$

$$= 44.38 \text{ ug/ml}$$

Step 2: Applying Dilution Factor

Dilution factor = 5.9

Actual concentration = 44.38×5.9

$$= 261.84 \text{ ug/ml}$$

Step 3: Calculation for 350 mL Beverage

Total caffeine = 261.84×350

$$= 91644 \text{ ug}$$

$$= 91.64 \text{ mg}$$

The caffeine content in 350 mL of Monster drink was found to be: **91.64 mg**



Fig.no.12 - Monster Energy

Sample 5 – Hell

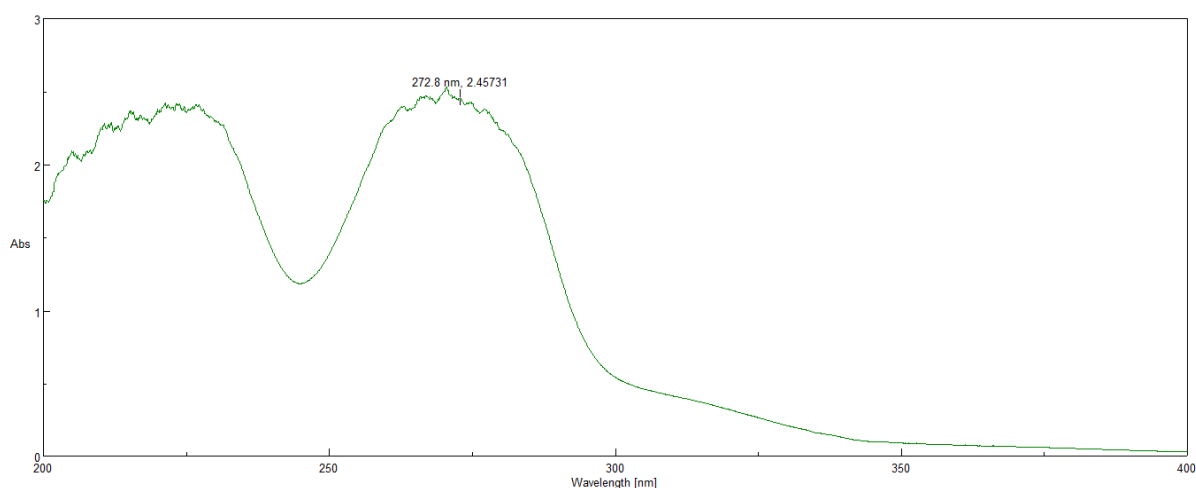


Fig.13 – Hell Energy Drink Spectra

Step 1: Substitution of Values

Absorbance of Hell energy drink = 2.4

$$X = 2.4 - 0.4756/0.0366$$

$$= 1.9244 / 0.0366$$

$$= 52.58 \text{ ug / ml}$$

Step 2: Applying Dilution Factor

Dilution factor = 6

$$\text{Actual concentration} = 52.58 \times 250$$

$$= 78870 \text{ ug}$$

$$= 78.87 \text{ mg}$$



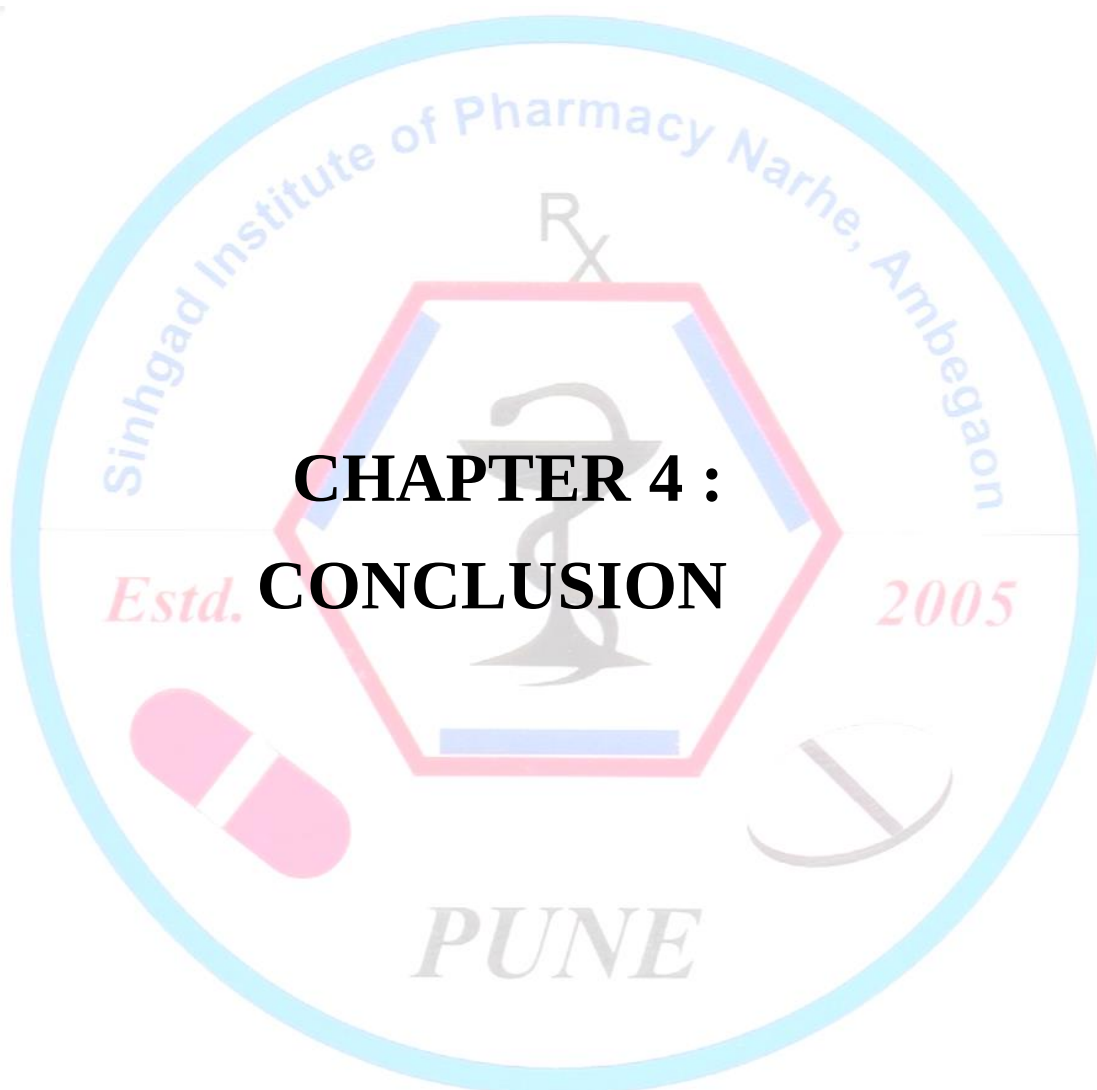
The caffeine content in 250 mL of Hell Energy Drink was found to be: **78.87 mg**

Fig .14 - Hell

Table 3

Sr. no.	Beverage Name	Volume (mL)	Absorbance	Dilution Factor	Caffeine Content (mg)	Label Claim (mg)
1	Sting Drink	300	2.30	5.75	71.66	72
2	Mountain Dew	250	1.9	3.25	50.61	52
3	Predator Drink	400	1.9	6	70.07	90
4	Monster Drink	350	2.1	5.9	91.64	90
5	Hell Energy Drink	250	2.4	6	78.87	75

The caffeine content in five different beverages was successfully determined using UV spectrophotometry. The method showed good linearity and reliability based on the calibration curve. Among the analyzed samples, Monster drink exhibited the highest caffeine content, while Mountain Dew showed the lowest caffeine content.



Conclusion

The present project entitled **“Quantitative Estimation and Comparative Study of Caffeine Content in Selected Beverages Using UV Spectrophotometry”** was successfully carried out with the objective of estimating and comparing the caffeine content present in different commercially available beverages.

In this study, different beverage samples were collected and analysed using standard UV spectrophotometric procedures. The calibration curve of caffeine was prepared by using standard caffeine solutions, and the absorbance values were measured at the selected wavelength. The obtained calibration curve showed good linearity, indicating that Beer-Lambert's law was followed within the selected concentration range. The caffeine content present in various beverages was then determined using the developed analytical method.

The experimental findings demonstrated that different beverages contain varying amounts of caffeine depending on their formulation, ingredients, and manufacturing processes. Energy drinks and coffee-based beverages were found to contain comparatively higher amounts of caffeine, whereas some soft drinks contained lower concentrations. The study also highlighted the importance of quality control and analytical evaluation in pharmaceutical and food industries to ensure consumer safety and proper labelling of caffeine-containing products.

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